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International Space Station TRACKER

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The proposed International Space Station (ISS) tracker, powered by the NodeMCU board, uses LEDs and a buzzer to indicate real-time ISS position.

The ISS tracker interfaces with the NASA API to access the ISS's current location data. Storing the user's longitude and latitude in the code, the device calculates the effective distance between the ISS and the user's location. Utilising LEDs and a buzzer, the system effectively indicates the ISS's distance. A formula checks if the ISS is overhead. The green LED lights up when the ISS is directly overhead, and red LEDs light up progressively as the ISS nears the user's location. The author's prototype is shown in Fig. 1.

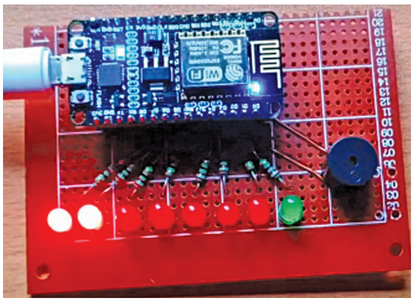

Fig. 1: Author's prototype

Communication with astronauts or others in the vicinity is possible when the ISS passes over, assuming they are on the radio at that time. This communication, facilitated by the ISS's radio repeater, requires an antenna. The components needed to build the tracker are listed in bill of materials table.

BILL OF MATERIALS

Components	Quantity
NodeMCU (MOD1)	1
Red LEDs (LED1-LED7)	7
Green LED (LED8)	1
Resistor 220Ω (R1-R8)	8
Buzzer	1

The circuit diagram of the ISS tracker is shown in Fig. 2. The tracker is built around NodeMCU (MOD1), a green LED (LED1), seven red LEDs (LED2 through LED8), and eight 220-ohm resistors (R1 through R8).

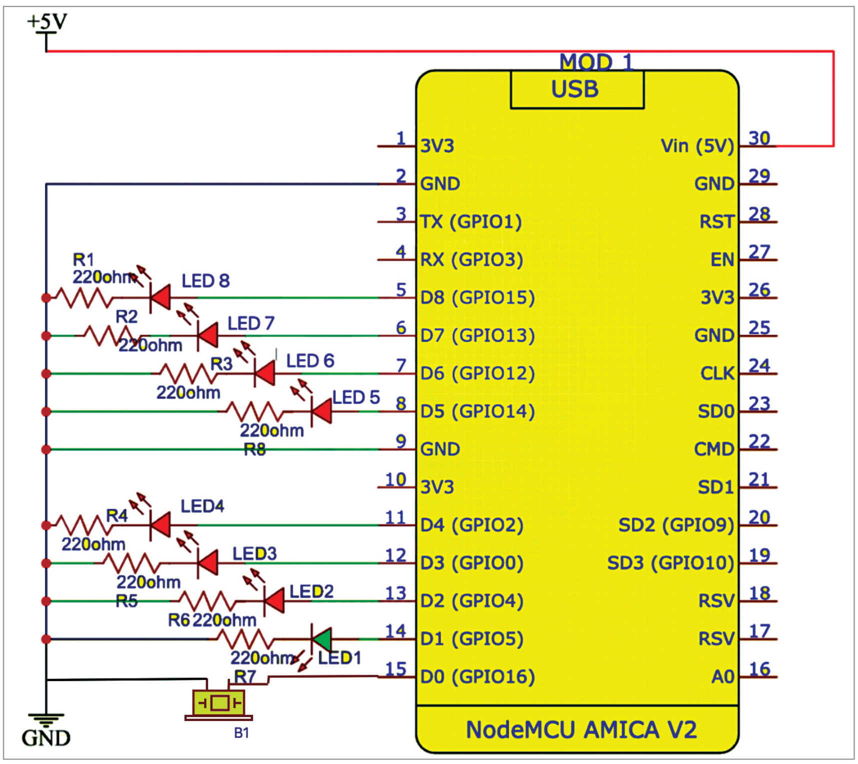

Fig. 2: Circuit diagram

Software

The code is prepared using Arduino IDE. In the first part of the code, the library for Wi-Fi configuration is added, and then the Wi-Fi credentials are set in the code, such as the SSID and password of the Wi-Fi network.

The server URL is set for obtaining location of the ISS. In Fig. 3, the code snippet shows the Wi-Fi credentials, while Fig. 4 demonstrates the loop function acquiring the ISS distance.

Next, the loop function is set up to extract the distance of the ISS tracker and the LEDs (LED2 through LED8) glow according to the distance from the device location. When the ISS comes closest to the device, the green LED1 starts glowing. Also buzzer sounds when the ISS comes just above the tracker.

Construction and testing

First, select the correct port and board in the Arduino IDE and upload